

Literature Survey: 2D Systolic Array-Based Processing Elements for Matrix-Vector/Matrix-Matrix Multiplication Acceleration

Domain: Deep Learning, Digital Signal Processing, and Scientific Computing

Date: September 2026

Scope: IEEE/ACM/Springer publications (2014–2025), plus foundational works

Papers Surveyed: 22 (3 foundational + 19 recent)

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1. Introduction

Matrix-vector and matrix-matrix multiplication form the computational backbone of deep learning inference and training, digital signal processing (DSP) pipelines, and scientific computing workloads. The systolic array architecture, originally proposed by H. T. Kung and C. E. Leiserson in 1978, provides a highly regular, pipelined, and locally connected array of processing elements (PEs) that rhythmically pass data through a network—maximizing data reuse and minimizing memory bandwidth requirements.

Over the past decade, the systolic array has been revitalized as the core computation engine in modern accelerators, most notably Google's Tensor Processing Unit (TPU). This survey examines **22 research papers** spanning foundational works, ASIC implementations, FPGA-based designs, and simulation-based architectural explorations. For each paper, we extract bibliographic information, architecture details, resource utilization metrics, performance metrics, stated limitations, and comparative positioning against prior systolic array designs.

2. Foundational Works

Paper F1: Systolic Arrays (for VLSI)

Field	Details
Authors	H. T. Kung, C. E. Leiserson
Year	1978/1979
Venue	Sparse Matrix Symposium / CMU Technical Report CMU-CS-79-103

Architecture Summary: Introduced the concept of "systolic" computation. Demonstrated hexagonally connected processor networks for matrix multiplication and LU decomposition. PE design: Simple MAC units with nearest-neighbor communication.

Key Contribution: Coined the term "systolic" and established foundational principles of regular, local, and pipelined computation for VLSI.

Stated Limitations: Limited to compute-bound problems; theoretical VLSI feasibility without silicon implementation.

Paper F2: Why Systolic Architectures?

Field	Details
Authors	H. T. Kung
Year	1982
Venue	<i>Computer</i> (IEEE), Vol. 15, No. 1, pp. 37-46
DOI	10.1109/MC.1982.1653825

Key Contribution: Provided the definitive justification for systolic architectures as a scalable solution for compute-intensive applications.

Paper F3: Eyeriss — Energy-Efficient Dataflow for CNNs

Field	Details
Authors	Y.-H. Chen, T. Krishna, J. Emer, V. Sze
Year	2016
Venue	ISCA 2016 / ISSCC 2016
DOI	10.1109/ISCA.2016.40
Array Size	12×14 (168 PEs)
PE Design	Row-Stationary (RS)
Platform	ASIC, 65 nm CMOS
Frequency	200 MHz
Power	~278 mW
Area	12.25 mm ²

3. Detailed Paper Survey

Paper 1: TPU v1 — In-Datcenter Performance Analysis

Field	Details
Authors	N. P. Jouppi et al.
Year / Venue	2017 / ISCA 2017
DOI	10.1145/3079856.3080246
Array Size	256×256 (65,536 MACs)
PE Design	Weight-Stationary (WS)
Platform / Node	ASIC / 28 nm
Clock Frequency	700 MHz
Peak Throughput	92 TOPS (INT8)
Power	40 W (TDP 75 W)
Energy Efficiency	~2.3 TOPS/W
Die Area	331 mm ²
On-chip Memory	24 MiB Unified Buffer + 4 MiB accumulators

Limitations: Inference-only; INT8-only; DDR3 memory bandwidth bottleneck; fixed 256×256 array → underutilization; no sparsity support.

Comparative: 15–30× faster than CPUs/GPUs; 30–80× better energy efficiency.

Paper 2: TPU v4 — Optically Reconfigurable Supercomputer

Field	Details
Authors	N. P. Jouppi et al.
Year / Venue	2023 / ISCA 2023
DOI	10.1145/3579371.3589350
Array Size	128×128 MXUs (×4 per chip)
PE Design	Weight-Stationary (WS); bf16 multiply, FP32 accumulate
Platform / Node	ASIC / 7 nm
Clock Frequency	1050 MHz
Peak Throughput	275 TFLOPS (bf16/INT8) per chip
Power	170 W (mean); 192 W (max)
Energy Efficiency	2.7× improvement over TPU v3 per watt
HBM	32 GiB HBM2, 1200 GB/s
Pod Scale	4096 chips → 1.1 EFLOPS

Limitations: High system complexity/cost; SparseCores specialized, not general-purpose; per-chip power ~170 W.

Comparative: 2.1× faster per chip vs. TPU v3; 10× faster at supercomputer scale; 2–3× more energy-efficient than contemporary ML DSAs.

Paper 3: Eyeriss v2 — Flexible Accelerator for Mobile DNNs

Field	Details
Authors	Y.-H. Chen, T.-J. Yang, J. Emer, V. Sze
Year / Venue	2019 / IEEE JSSC / ISSCC 2019
DOI	10.1109/JSSC.2018.2886423
PE Design	Row-Stationary Plus (RS+)
Platform / Node	ASIC / 65 nm
Throughput	1470.6 inferences/sec (MobileNet)
Energy Efficiency	2560.3 inferences/J
Improvement	12.6× speed, 2.5× energy over Eyeriss v1

Limitations: 65nm limits performance; PE area 75% scratchpad memory; complex control; inference-only.

Paper 4: MAERI — Reconfigurable Interconnects (ASPLOS 2018)

Field	Details
Authors	H. Kwon, A. Samajdar, T. Krishna
Year / Venue	2018 / ASPLOS 2018
DOI	10.1145/3173162.3173176
PE Design	Flexible (any dataflow via programmable switches)
Platform	Simulation

Limitations: Programmable interconnects add area/power overhead; routing complexity; no silicon validation.

Paper 5: SIGMA — Sparse & Irregular GEMM Accelerator (HPCA 2020)

Field	Details
Authors	E. Qin, A. Samajdar, H. Kwon et al.
Year / Venue	2020 / HPCA 2020
DOI	10.1109/HPCA47549.2020.00015
PE Design	Flex-DPE; Benes network $O(1)$ distribution; FAN $O(\log_2 N)$ reduction
Performance	5.7× vs. systolic (sparse); 3× vs. prior sparse accelerators

Limitations: Complex Benes interconnect area overhead; no silicon validation; interconnect power not fully characterized.

Paper 6: Gemmini — Full-Stack DNN Architecture Evaluation (DAC 2021 Best Paper)

Field	Details
Authors	H. Genc, S. Kim, A. Prakash et al.
Year / Venue	2021 / DAC 2021
DOI	10.1109/DAC18074.2021.9586216
PE Design	OS / WS (runtime selectable); configurable 16×16 mesh
Platform / Node	ASIC (RISC-V RoCC) / 16nm, 22FFL
Speedup	Up to 1000× vs. CPU

Limitations: Config-dependent performance; RISC-V only; no built-in sparsity support.

Paper 7: SCALE-Sim — Systolic CNN Accelerator Simulator (ISPASS 2020)

Field	Details
Authors	A. Samajdar, Y. Zhu, P. N. Whatmough et al.
Year / Venue	2018/2019 / ISPASS 2020
DOI	10.1109/ISPASS48437.2020.00028
Type	Cycle-accurate simulator; supports WS/OS/IS dataflows

Limitations: Ideal PE assumption; no sparsity in early versions; accuracy depends on memory model.

Papers 8–22: Summary Table

#	Name	Year	Key Innovation	Key Metric
8	VerSA	2023/24	Dense+sparse via bypass paths (256×256)	1.21×–1.60× sparse gain; 0.52% dense overhead
9	VUSA	2025	Virtual upscaling for unstructured sparsity (16nm)	37% area reduction; 68% power efficiency gain
10	PI-SA	2023	Fine-grained MAC power gating (32×32, 64×64)	32% area, 57% power reduction vs. Gemmini
11	Pruning Co-design	2024	Block pruning matched to SA mesh dimensions	Hardware-native zero skipping
12	Sense	2022/23	Channel clustering, load-balanced pruning	Significant DRAM reduction vs. Swallow
13	DSP Multiplexing	2023	In-DSP multiplexing, ring accumulators (FPGA)	20%–85% LUT reduction; 200–600+ MHz
14	BitSys	2023/24	Bit-slice reconfigurable INT4/INT8 (FPGA)	1.3×–3.5× speedup; 4–5 TOPS/W
15	ReNPU	2024	Unified datapath INT4/INT8/FP4/FP8/Posit	Multi-format support for mobile SoC
16	2D Knapsack	2022/23	Parallelism-aware SA partitioning (FPGA)	7×–22.7× throughput improvement
17	SAPER-AI	2023	Coarse row/column power gating (32×32, 64×64)	10%–25% power efficiency improvement
18	C-SIMD	2023	CORDIC-based configurable SIMD PE	14.29% area savings; 4× perf-efficiency
19	Approx. Computing	2022/23	Approximate multiplier-based MAC	20%–35% energy; 20%–25% area savings
20	HTMR	2023	Heterogeneous TMR fault tolerance	Up to 50% fault rate tolerance
21	Factored SA	2022	Radix-8 multiplier-based PE	Improved delay, area, power
22	DA Systolic	2022	Distributed arithmetic for low-power CNN	Reduced power and complexity

4. Comprehensive Comparison Tables

Table A: Architecture Comparison

#	Paper	Year	Array Size	Dataflow	Platform	Node
F1	Kung & Leiserson	1978	Theoretical	General	Theory	N/A
F2	Kung	1982	Theoretical	General	Theory	N/A
F3	Eyeriss v1	2016	12×14 (168)	Row-Stationary	ASIC	65nm
1	TPU v1	2017	256×256	Weight-Stationary	ASIC	28nm
2	TPU v4	2023	128×128 (×4)	Weight-Stationary	ASIC	7nm
3	Eyeriss v2	2019	Configurable	RS+	ASIC	65nm
4	MAERI	2018	Configurable	Flexible	Sim	N/A
5	SIGMA	2020	Flex-DPE	No-local-reuse	Sim	N/A
6	Gemmini	2021	Config (16×16)	OS/WS select	ASIC	16nm
8	VerSA	2023	256×256	Modified WS	Sim	N/A
9	VUSA	2025	Virtual	Modified Systolic	ASIC	16nm
10	PI-SA	2023	32×32 / 64×64	WS + Gating	ASIC	N/S
14	BitSys	2023	Configurable	Bit-slice	FPGA	N/A

Table B: Performance Metrics Comparison

#	Paper	Freq.	Throughput	Power	Energy Eff.	Area
F3	Eyeriss v1	200 MHz	N/S	~278 mW	Optimized	12.25 mm ²
1	TPU v1	700 MHz	92 TOPS	40 W	~2.3 TOPS/W	331 mm ²
2	TPU v4	1050 MHz	275 TFLOPS	170 W	2.7× vs v3	N/S
3	Eyeriss v2	N/S	1470.6 inf/s	N/S	2560.3 inf/J	N/S
5	SIGMA	N/S	5.7× vs SA	N/S	3× vs sparse	N/S
6	Gemmini	N/S	1000× vs CPU	N/S	N/S	N/S
8	VerSA	N/S	1.2–1.6× sparse	N/S	7.5–30.2% sav.	N/S
9	VUSA	N/S	N/S	68% impr.	68% gain	37% reduct.
10	PI-SA	N/S	N/S	57% reduct.	10–25% impr.	32% reduct.
14	BitSys	N/S	1.3–3.5× speed	N/S	4–5 TOPS/W	1.5–2× LUT red.

N/S = Not Specified in the paper

5. Synthesis and Analysis

5.1 Performance Trends

1. **Throughput scaling with technology nodes:** TPU v1 (28nm, 92 TOPS) → TPU v4 (7nm, 275 TFLOPS) demonstrates ~3× throughput improvement over 6 years.
2. **Frequency scaling:** ASIC: 200 MHz (65nm) → 700 MHz (28nm) → 1050 MHz (7nm). FPGA: 200–600 MHz.
3. **Sparse acceleration:** 1.2×–5.7× improvement. VerSA: 1.21×–1.60× with 0.52% dense overhead. SIGMA: 5.7× with higher complexity.
4. **Mixed-precision gains:** BitSys achieves 1.3×–3.5× throughput via INT4/INT8 switching.
5. **Supercomputer scale:** TPU v4 pods (4096 chips) achieve 1.1 EFLOPS.

5.2 Power/Energy Efficiency Trends

1. **ASIC evolution:** Eyeriss (~278 mW) → TPU v1 (~2.3 TOPS/W) → TPU v4 (2.7× vs v3) → Industry mobile (>11 TOPS/W).
2. **Power gating:** PI-SA: up to 57% power reduction. SAPER-AI: 10%–25% efficiency gain.
3. **Approximate computing:** 20%–35% energy savings (inference-only, error-resilient apps).
4. **Sparsity-driven:** VUSA: 68% power efficiency gain. VerSA: 7.5%–30.2% energy savings.
5. **FPGA-specific:** 4–5 TOPS/W via mixed-precision and DSP packing.

5.3 Hardware Resource Trends (LUT/DSP/FF/Transistor)

1. **ASIC area:** TPU v1: 331 mm² (28nm); Eyeriss v1: 12.25 mm² (65nm); scratchpad dominates (~75% of PE area).
2. **FPGA optimization:** LUT reductions 20%–85%; DSP efficiency +30% via packing; hybrid DSP/LUT architectures.
3. **Area reduction techniques:** Power gating: 32%; Sparsity: 37%; Approximate: 20%–25%; CORDIC: 14.29%.
4. **Memory dominance:** On-chip memory exceeds 50–75% of total area across all designs.

5.4 Common Limitations Across Papers

Limitation	Papers Reporting	Frequency
PE underutilization at boundary conditions	TPU v1, SIGMA, VerSA, Pruning Co-design, SAPER-AI	Very High
Fixed array dimensions ("square curse")	TPU v1, SIGMA, MAERI, SCALE-Sim, Boundary studies	Very High
Memory bandwidth bottleneck	TPU v1, TPU v4, FPGA designs, SCALE-Sim	High
Sparsity handling gaps	TPU v1/v4, Eyeriss v1, VerSA, Sense	High
Scalability constraints	MAERI, SIGMA, FPGA designs, PI-SA	High
Precision/quantization trade-offs	BitSys, ReNPU, Approx. Computing, TPU v1	Moderate
Control/synchronization overhead	Eyeriss v1/v2, MAERI, SIGMA, VerSA	Moderate
Data reuse inefficiency	SCALE-Sim analysis, FPGA designs	Moderate
Lack of silicon validation	MAERI, SIGMA, VerSA, SCALE-Sim, Pruning Co-design	Moderate
Thermal/power constraints at scale	TPU v4, large FPGA designs	Moderate

5.5 Identified Research Gaps

1. **Scalable, Dimension-Adaptive Systolic Arrays:** Most designs use fixed $N \times N$ arrays. Dynamic reconfiguration to match workload shapes without overhead remains underexplored.
2. **Unified Dense-Sparse Processing:** No design achieves near-optimal utilization for both without significant area/complexity overhead.
3. **Cross-Domain Versatility:** DSP and scientific computing applications remain underexplored.
4. **Energy-Proportional Computing:** Power scaling linearly with useful computation is an open problem.
5. **Standardized Benchmarking:** No unified framework for fair cross-architecture comparison.
6. **Boundary PE Optimization:** Widely acknowledged but insufficiently addressed.
7. **Memory-Compute Co-optimization:** On-chip memory organization remains largely ad-hoc.
8. **Fault Tolerance with Low Overhead:** Lightweight built-in resilience for edge deployment is underdeveloped.

6. Conclusion: Motivating a New Scalable 2D Systolic Array Design

This comprehensive survey of 22 research papers reveals that while 2D systolic arrays have achieved remarkable success as the computational backbone of modern deep learning accelerators—culminating in designs like the TPU v4 delivering 275 TFLOPS at 7nm—several fundamental limitations remain stubbornly unresolved across the literature.

The most frequently unresolved limitation is the rigid, fixed-dimension nature of conventional 2D systolic arrays, which leads to systematic PE underutilization at boundary conditions. When matrix/tensor dimensions do not evenly divide into the fixed $N \times N$ array dimensions, boundary PEs remain idle, wasting both area and energy. This "square curse" is explicitly acknowledged in papers spanning from the original TPU v1 (2017) through SIGMA (2020), MAERI (2018), and recent 2023–2024 scalability studies. Despite numerous proposed solutions—including flexible interconnects (MAERI, SIGMA), virtual upscaling (VUSA), and partitioning strategies (2D Knapsack)—each introduces its own overhead in the form of complex interconnects, scheduling logic, or sub-optimal resource utilization.

Closely coupled with this is the inability of most systolic arrays to efficiently handle both dense and sparse workloads without either (a) dedicating separate hardware for each regime, or (b) accepting significant overhead for the flexibility. VerSA achieves a promising 0.52% dense overhead with $1.21\times$ – $1.60\times$ sparse improvement, but this remains below the $5.7\times$ improvement of dedicated sparse accelerators like SIGMA.

A third persistent gap is the lack of energy-proportional operation: most systolic arrays consume near-constant power regardless of actual PE utilization, and fine-grained power management adds control complexity that partially offsets efficiency gains.

These converging limitations strongly motivate the development of a **new scalable 2D systolic array design** that addresses: (1) dynamic dimension adaptability to eliminate boundary PE waste, (2) unified dense-sparse processing within the same regular array structure, and (3) inherently energy-proportional operation that scales power consumption with actual computational workload—all while maintaining the fundamental simplicity, regularity, and local communication properties that make systolic arrays uniquely suitable for VLSI implementation.

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